ASSESSMENT TASK 2

Data-Driven Customer Segmentation for Strategic Marketing in Fitness Centres

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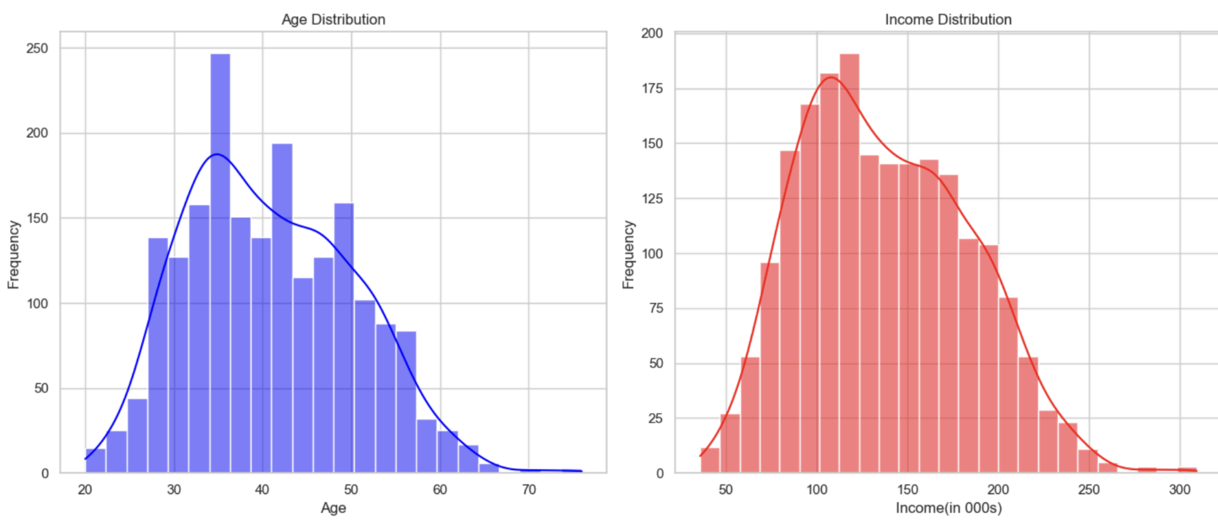
Introduction

The fitness industry is intensely competitive, necessitating gyms to comprehend their clientele to enhance engagement, retention, and profitability. This report provides a data-driven segmentation analysis of 2,000 gym members, utilising demographic and socioeconomic characteristics including age, income, gender, education, occupation, and settlement size. The goal is to identify actionable customer segments for targeted marketing and service design. The methodology encompassed exploratory data analysis, succeeded by clustering through K-means++ and Agglomerative Clustering, with results analysed and contrasted to guarantee robustness.

Exploratory Data Analysis

<u>1. Age and Income:</u>



The age distribution exhibits a slight right skew, signifying a higher concentration of younger to middle-aged clientele. A gradual decrease in frequency occurs after the age of 50.

Income demonstrates a pronounced positive skew, with the majority of individuals earning between $80,000 and $180,000. The long tail emphasises a minor high-income demographic surpassing $250,000, potentially comprising executives or wealthy professionals.

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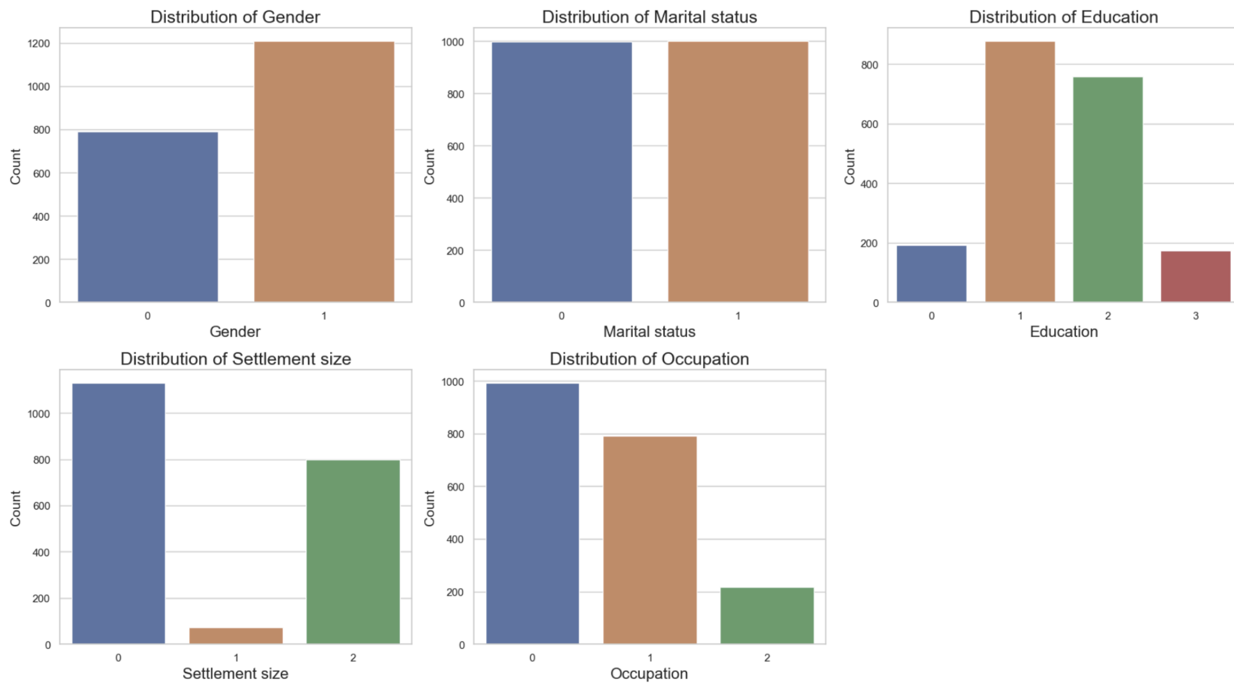
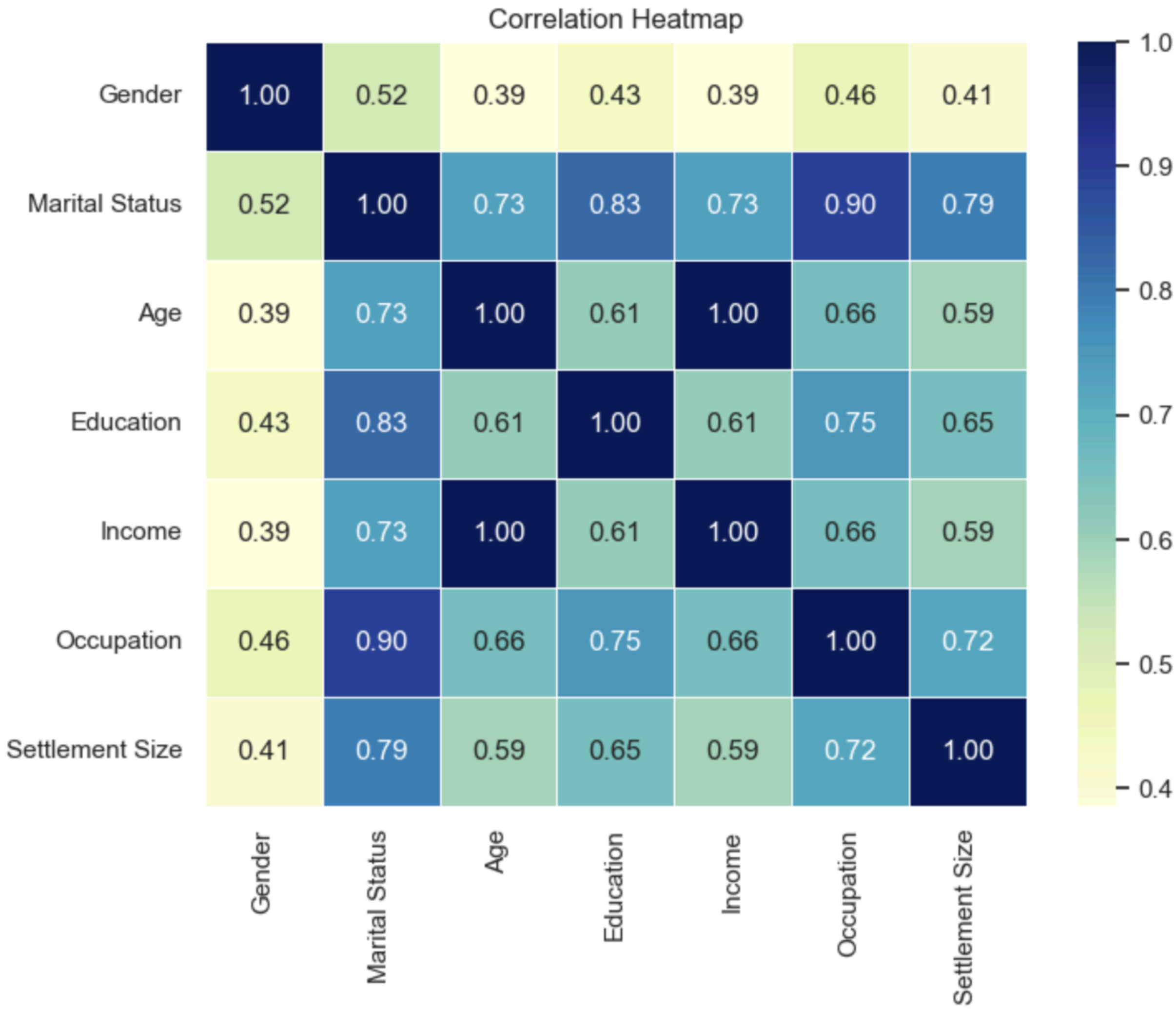
<u>2.Bar Charts: Categorical Variables</u>



The dataset indicates a modest male predominance and an equitable distribution of marital status, with the majority of participants possessing tertiary education and employed in entry to mid-level positions. Settlement patterns are bimodal—either urban or rural—emphasizing dual market potential. These factors indicate diverse lifestyles that are likely to affect gym utilisation and service preferences.

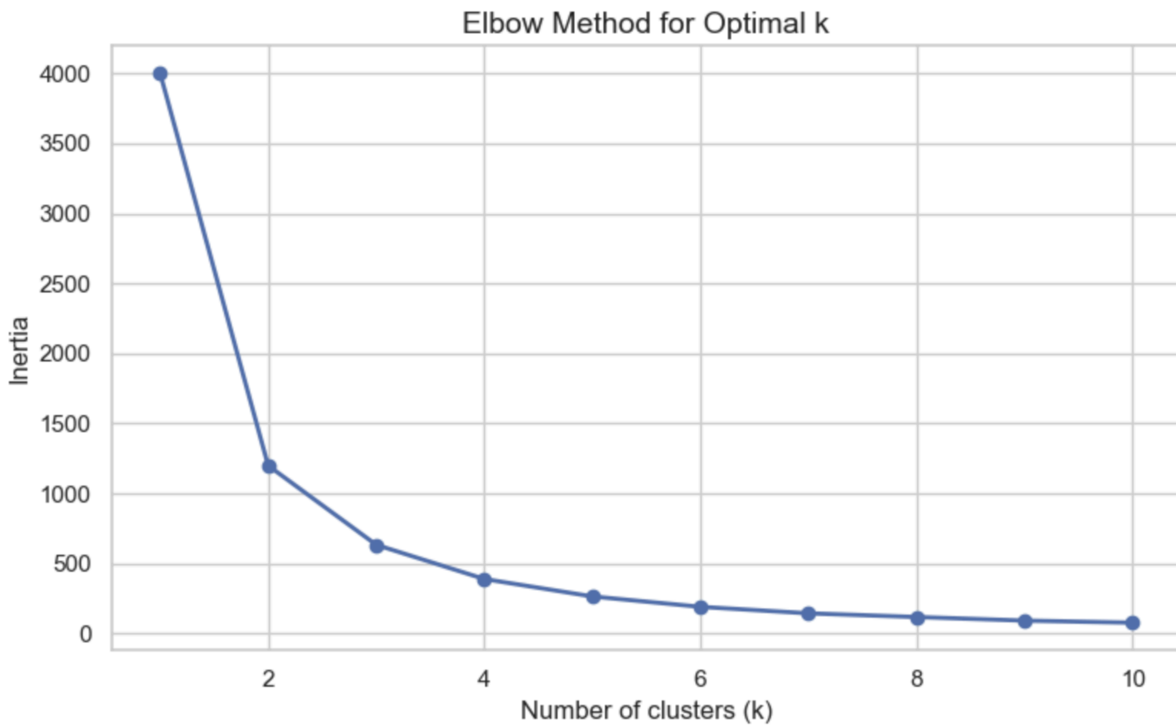
3.Correlation Matrix Visualisation



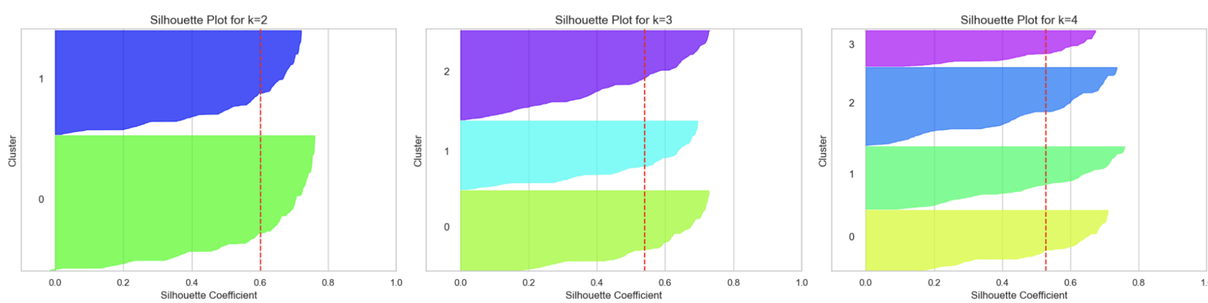
A moderate positive correlation between age and income indicates career advancement as earnings rise with age.
Education and income are linked, confirming the link between education and income.
Urbanites earn more due to the concentration of jobs in cities, as income and settlement size are positively correlated.
Additionally, occupation and education have a moderate correlation, reinforcing rational socioeconomic mobility.

Customer Segmentation:

<u>Elbow Method:</u>



<u>Silhouette Analysis:</u>



Two established methodologies were employed to ascertain the optimal quantity of customer segments and both methodologies strongly suggested that three customer clusters represented the most effective segmentation.

K-means++ Clustering

Cluster	Gender	Marital Status	Age (yrs)	Education	Income ($)	Occupation	Settlement Size	Count
0	0.39	0.07	31.5	0.90	91,820	0.09	0.16	778
1	0.84	0.97	53.6	2.08	200,089	1.18	1.58	494
2	0.67	0.64	42.1	1.63	143,891	0.78	1.04	728

Three separate clusters were identified, comprising customer counts of 989, 798, and 213. Each cluster exhibited distinct variations in age, income, education, and other attributes.

Agglomerative Clustering

Cluster	Gender	Marital Status	Age (yrs)	Education	Income ($)	Occupation	Settlement Size	Count
0	0.67	0.64	42.5	1.63	145,633	0.78	1.04	969
1	0.38	0.05	30.7	0.87	88,085	0.07	0.13	668
2	0.85	0.98	55.5	2.10	209,085	1.18	1.59	350

Agglomerative Clustering similarly categorised customers into three distinct clusters of sizes 992, 798, and 210. The central values for each cluster exhibited consistent patterns across methodologies, thereby reinforcing the reliability of the findings.

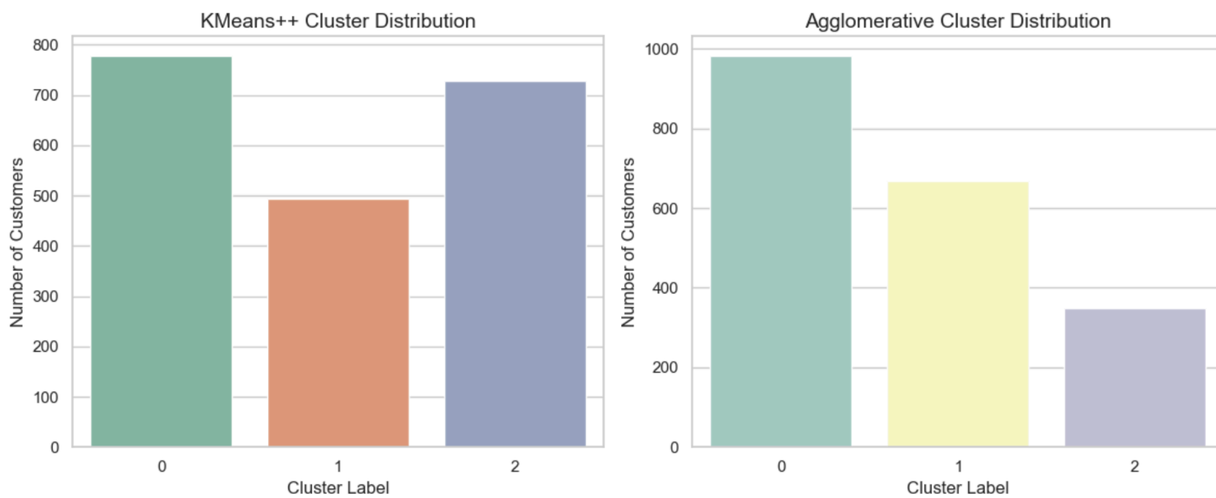
Segment Interpretation and Naming :

This section analyses the customer segments identified through K-means++ and Agglomerative Clustering. Each segment is characterised by essential demographic and behavioural attributes: age, income, gender, education, marital status, occupation, and type of location .

Segments:

The charts below illustrates the quantity of customers in each cluster generated by the two methodologies:

Figure 6: Distribution of Clusters by Clustering Methodology



K-means++ generated clusters with sizes of 778, 494, and 728, respectively.

Agglomerative Clustering yielded comparable clusters: 798, 668, and 350 customers.

The Clusters:

Cluster 0 – "Youthful, Economically-Conscious Individuals"
Young, unmarried men in rural or small-town areas with low income and education. They are likely students or early-career professionals. This cost-sensitive, digitally engaged demographic is ideal for low-cost memberships, flexible plans, and online promotions..

Cluster 1 – "Wealthy Metropolitan Executives"
Metropolitan-based older, married women with high income and education. This lucrative segment responds well to premium services like personal training, wellness programmes, and loyalty programmes.

Cluster 2 – "Knowledgeable Suburban Adults"
Middle-aged suburbanites with higher education and moderate to high incomes. They value quality and value despite being financially secure. Ideal for family packages, flexible scheduling, and community services.

Cluster Overlap Matrix:

A cross-tabulation matrix was constructed to assess the categorisation of customers across both methods.

Table 3: Intersection of Customer Allocations

K-means++ Clustering	Agglo Cluster 0	Agglo Cluster 1	Agglo Cluster 2	Optimal Correspondence
Cluster 0	110	668	0	Agglo Cluster 1 → Budget-Conscious
Cluster 1	144	0	350	Agglo Cluster 2 → Metropolitan Executives
Cluster 2	728	0	0	Agglo Cluster 0 → Suburban Adults

The significant overlap in cluster membership verifies that both algorithms are identifying consistent behavioural and demographic patterns rather than random groupings.
These segments furnish the gym chain with a data-driven basis for:

- Customised product offerings
- Focused promotional initiatives
- Strategic allocation of resources in services

The segmentation is both robust across methodologies and aligned with distinct lifestyle and income disparities, rendering it highly effective for customer engagement and retention strategies.

Marketing Recommendations Based on Customer Segments Identified

Group 1: Youthful, Economically-Conscious Individuals

Gamified Membership Programme
Propose a digital plan priced at $5–$7 per week, incorporating incentives for exercise and referrals to enhance motivation and foster loyalty among cost-sensitive users.

Collective Hustle Permit
Implement off-peak group memberships for 2–4 individuals, promoting participation via shared expenses and social responsibility.

University Fitness Competition
Collaborate with student organisations to implement app-based challenges offering gym rewards, creating excitement and drawing in new users without utilising discounts.

Group 2: Wealthy Metropolitan Executives

Intelligent Wellness Strategy
Combine gym access with monthly coaching and health monitoring to promote sustained wellbeing with minimal time commitment.

Dual Flex Couples Plan
Establish a versatile membership for two individuals featuring hybrid access and scheduling, suitable for busy households or couples.

Community Loyalty Programme
Incentivize attendance, referrals, and event participation via a points-based system that fosters habitual engagement and brand affinity.

Group 3: Knowledgeable Suburban Adults

Executive Wellness Package
Provide premium memberships that include personal training, nutrition, massage, and concierge services for affluent, time-constrained clients.

Corporate Wellness Collaborations
Engage with employers to offer exclusive membership benefits, on-site courses, and yearly health incentives.

Exclusive Women's Wellness Programme
Implement customised wellness services for women, encompassing coaching, nutritional consultations, and exclusive studio experiences to enhance loyalty.

Conclusion:

This report categorised 2,000 gym members into three distinct groups utilising demographic and behavioural data. K-means++ and Agglomerative Clustering revealed consistent patterns, enhancing reliability. Each segment was analysed and aligned with specific marketing strategies. These insights allow the gym to customize offerings, enhance customer engagement, and optimize resource allocation—transitioning from generic outreach to data-driven, segment-specific strategies that promote retention, satisfaction, and sustained business growth.